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Send emails easily in Python (with attachments and multipart)

<> email_utils.py

Raw

```
1  #!/usr/bin/env python
2  # coding: utf-8
3
4  # This little project is hosted at: <https://gist.github.com/1455741>
5  # Copyright 2011-2020 Álvaro Justen [alvarojusten at gmail dot com]
6  # License: GPL <http://www.gnu.org/copyleft/gpl.html>
7
8  import os
9  from email.mime.multipart import MIMEMultipart
10 from email.mime.text import MIMEText
11 from email.mime.base import MIMEBase
12 from mimetypes import guess_type
13 from email.encoders import encode_base64
14 from getpass import getpass
15 from smtplib import SMTP
16
17
18 def get_email(email):
19     if '<' in email:
20         data = email.split('<')
21         email = data[1].split('>')[0].strip()
22     return email.strip()
23
24 class Email(object):
25     def __init__(self, from_, to, subject, message, message_type='plain',
26                  attachments=None, cc=None, message_encoding='us-ascii'):
27         self.email = MIMEMultipart()
28         self.email['From'] = from_
29         self.email['To'] = to
30         self.email['Subject'] = subject
31         if cc is not None:
32             self.email['Cc'] = cc
33         text = MIMEText(message, message_type, message_encoding)
34         self.email.attach(text)
35         if attachments is not None:
36             for filename in attachments:
37                 mimetype, encoding = guess_type(filename)
38                 mimetype = mimetype.split('/', 1)
39                 fp = open(filename, 'rb')
40                 attachment = MIMEBase(mimetype[0], mimetype[1])
41                 attachment.set_payload(fp.read())
42                 fp.close()
43                 encode_base64(attachment)
44                 attachment.add_header('Content-Disposition', 'attachment',
45                                     filename=os.path.basename(filename))
46                 self.email.attach(attachment)
47
48     def __str__(self):
49         return self.email.as_string()
50
51
52 class EmailConnection(object):
53     def __init__(self, server, username, password):
54         if ':' in server:
55             data = server.split(':')
56             self.server = data[0]
57             self.port = int(data[1])
58         else:
```

```

59         self.server = server
60         self.port = 25
61         self.username = username
62         self.password = password
63         self.connect()
64
65     def connect(self):
66         self.connection = SMTP(self.server, self.port)
67         self.connection.ehlo()
68         self.connection.starttls()
69         self.connection.ehlo()
70         self.connection.login(self.username, self.password)
71
72     def send(self, message, from_=None, to=None):
73         if type(message) == str:
74             if from_ is None or to is None:
75                 raise ValueError('You need to specify `from_` and `to`')
76             else:
77                 from_ = get_email(from_)
78                 to = get_email(to)
79             else:
80                 from_ = message.email['From']
81                 if 'Cc' not in message.email:
82                     message.email['Cc'] = ''
83                 to_emails = [message.email['To']] + message.email['Cc'].split(',')
84                 to = [get_email(complete_email) for complete_email in to_emails]
85                 message = str(message)
86             return self.connection.sendmail(from_, to, message)
87
88     def close(self):
89         self.connection.close()

```

 [example_email_utils.py](#)

[Raw](#)

```

1  #!/usr/bin/env python
2  # coding: utf-8
3
4  # This script asks your name, email, password, SMTP server and destination
5  # name/email. It'll send an email with this script's code as attachment and
6  # with a plain-text message. You can also pass `message_type='html'` in
7  # `Email()` to send HTML emails instead of plain text.
8  # You need email_utils.py to run it correctly. You can get it on:
9  #         https://gist.github.com/1455741
10 # Copyright 2011-2020 Álvaro Justen [alvarojusten at gmail dot com]
11 # License: GPL <http://www.gnu.org/copyleft/gpl.html>
12
13 import sys
14 from getpass import getpass
15 from email_utils import EmailConnection, Email
16
17
18 print 'I need some information...'
19 name = raw_input(' - Your name: ')
20 email = raw_input(' - Your e-mail: ')
21 password = getpass(' - Your password: ')
22 mail_server = raw_input(' - Your mail server: ')
23 to_email = raw_input(' - Destination email: ')
24 to_name = raw_input(' - Name of destination: ')
25 subject = 'Sending mail easily with Python'
26 message = 'here is the message body'
27 attachments = [sys.argv[0]]
28
29 print 'Connecting to server...'
30 server = EmailConnection(mail_server, email, password)
31 print 'Preparing the email...'
32 email = Email(from_='%s' % (name, email), #you can pass only email
33              to='%s' % (to_name, to_email), #you can pass only email
34              subject=subject, message=message, attachments=attachments)
35 print 'Sending...'
36 server.send(email)
37 print 'Disconnecting...'
38 server.close()
39 print 'Done!'

```

 [send_mail.py](#)

[Raw](#)

```

1  #!/usr/bin/env python3

```

```

2 # Script to send emails using Python using the command-line.
3 # Copyright 2020 Álvaro Justen [alvarojusten at gmail dot com]
4 # License: GPL <http://www.gnu.org/copyleft/gpl.html>
5 import argparse
6 import os
7 import smtplib
8 import sys
9 from email.mime.multipart import MIMEMultipart
10 from email.mime.text import MIMEText
11
12
13 def send_mail(
14     smtp_host, smtp_port, smtp_user, smtp_pass, from_address, to_addresses,
15     subject, text, html=False
16 ):
17     # TODO: use `emails_utils` code
18     message = MIMEMultipart()
19     message["From"] = from_address
20     message["To"] = ",".join(to_addresses)
21     message["Subject"] = subject
22     if html:
23         message.attach(MIMEText(text, "html", "utf-8"))
24     else:
25         message.attach(MIMEText(text, "plain", "utf-8"))
26     server = smtplib.SMTP(host=smtp_host, port=smtp_port)
27     server.starttls()
28     server.login(smtp_user, smtp_pass)
29     server.sendmail(from_address, to_addresses, message.as_string())
30     server.quit()
31
32
33 if __name__ == "__main__":
34     parser = argparse.ArgumentParser()
35     parser.add_argument("--html", action="store_true")
36     parser.add_argument("--smtp-host", default=os.environ.get("SMTP_HOST"))
37     parser.add_argument("--smtp-port", default=os.environ.get("SMTP_PORT", 587))
38     parser.add_argument("--smtp-user", default=os.environ.get("SMTP_USER"))
39     parser.add_argument("--smtp-pass", default=os.environ.get("SMTP_PASS"))
40     parser.add_argument("from_address")
41     parser.add_argument("to_addresses", help="If more than one, separate by comma")
42     parser.add_argument("subject")
43     parser.add_argument("message")
44     args = parser.parse_args()
45
46     if None in (args.smtp_host, args.smtp_port, args.smtp_user, args.smtp_pass):
47         print("ERROR: missing SMTP configuration", file=sys.stderr)
48         exit(1)
49
50     to_addresses = [
51         address.strip()
52         for address in args.to_addresses.split(",")
53         if address.strip()
54     ]
55     send_mail(
56         smtp_host=args.smtp_host,
57         smtp_port=args.smtp_port,
58         smtp_user=args.smtp_user,
59         smtp_pass=args.smtp_pass,
60         from_address=args.from_address,
61         to_addresses=to_addresses,
62         subject=args.subject,
63         text=args.message,
64         html=args.html,
65     )

```

[Load earlier comments...](#)



remmi11 commented on Jan 20, 2017 • edited ▼



Im having a similar issue. Mine seems to be related to the attachment bc I can successfully send without an attachment.

Trying to change
 attachments = [sys.argv[0]]

```
to
attachments = 'C:/Time App/Time.xlsx'

others I have tried...
attachments = 'C:\Time App\Time.xlsx'
..
path = 'C:\Time App\Time.xlsx'
attachments = os.path.abspath(path)

Error:
Traceback (most recent call last):
File "", line 254, in run_nodebug
File "C:\Time App\scripts\example_email_utils.py", line 36, in
subject=subject, message=message, attachments=attachments)
File "C:\Time App\scripts\email_utils.py", line 39, in init
mimetype = mimetype.split('/', 1)
AttributeError: 'NoneType' object has no attribute 'split'
```



remmi11 commented on Jan 21, 2017 • edited ▾

I believe I have found my answer. The problem revolved around my lack of understanding of what
Line: 27 attachments = [sys.argv[0]] was requesting.

I have a better understanding now thanks to this question asked on the stacks.
<https://stackoverflow.com/questions/33773957/sys-argv0-meaning-in-python#>

Fantastic tool by the way!



kootenpv commented on Feb 19, 2017

You could also try a lib for it :) <https://kootenpv.github.io/2016-04-24-yagmail> Disclaimer: I'm the creator ^^



angelinahli commented on Aug 17, 2017

Small detail, but instead of `if cc is not None` you can just write `if cc` , etc. (None values are falsy)



BeyMelamed commented on Dec 24, 2017

I found that trying to send email to multiple recipients did not work (only the first recipient got the message)
I tried this with comma delimited recipients as in: 'user1@gmail.com,user2@yahoo.com'
When i changed line 83 in EmailConnection class:
to_emails = [message.email['To']] + message.email['Cc'].split(',')
To:
to_emails = message.email['To'].split(',') + message.email['Cc'].split(',')
All recipients got the email



Macthalin commented on Feb 23, 2018

How to get rid of "no module named email.mime" problem?



Macthalin commented on Feb 23, 2018

What package should be installed for sending attachments through mail in pi3? And how to import?



rodrigoestevao commented on Nov 22, 2018

It was exactly what I was looking for, thank you!!!



pythonmars commented on Dec 31, 2018

When I run example_email_utils.py from Terminal (VS Code):

ImportError: cannot import name 'EmailConnection' from 'email_utils' (email_utils.py and example_email_utils.py are in the same directory).

pylint shows 2 problems:

1. No name 'EmailConnection' in module 'email_utils'
2. No name 'Email' in module 'email_utils'

Why is like that? Could anyone help me with this?

Thanks!



pythonmars commented [on Dec 31, 2018](#) ...

I solved the problem. It's running perfectly.
Only one warning in the class **Email(object)** at line **mimetype, encoding = guess_type(filename)**: *Unused variable 'encoding'*



pythonmars commented [on Dec 31, 2018](#) ...

ok, i found the answer:
change line mimetype, encoding = guess_type(filename) to:
mimetype, _ = guess_type(filename)
then no more warning



JachiOnuoha commented [on Jan 10, 2019](#) ...

Great Script!! But, how can I make the program include my gmail signature for outgoing emails?



prasathn1984 commented [on Nov 20, 2019](#) ...

Can anyone explain, how to attach recent backup file to send through mail by daily basis?



prasathn1984 commented [on Nov 20, 2019](#) ...

I am doing backupjob of my cisco devices and it will store with today's name. I want to attach and send mail of latest files to DL. How do i write script as i am new to python. Could you help me on this?



ayyoubmaulana commented [on Nov 20, 2019](#) via email ...

sorry can you introduce yourself first please ?

...



prasathn1984 commented [on Nov 20, 2019](#) via email ...

Hi

This is Prasath from India, I am new to python and writing script to take backup of my Cisco devices. It is working and storing all device backup with zip with respective date. I want to know how to attach latest zip file and send through mail.

On Wed, 20 Nov 2019, 5:18 pm ayyoubmaulana, <notifications@github.com> wrote:

...



advaybajaj commented [on Aug 2, 2020](#) ...

can someone please tell me what to write when we input "mail_server"



Sankalp004 commented [on Oct 28, 2020](#) • edited ...

can someone please tell me what to write when we input "mail_server"

SMTP Server Links :
Gmail-> smtp.gmail.com:587
Yahoo-> smtp.mail.yahoo.com:587
Outlook -> smtp-mail.outlook.com:587



SandhyaArja commented on [Sep 12, 2022](#)

```
hi, Im new to python script help to attach pdf in my code with the helpof encoders
from email.message import EmailMessage
import ssl ,smtplib
from pandas import *
df = pandas.read_csv("mail.csv")
f=df["Name"].tolist()
g=df["Pdf"].tolist()
print(g)
receiver=df["Mail"].tolist()
for f in receiver:
sender = "
paswd = "
subject = 'Going To Ice Cream'
body = 'Guys when we are going to have Fun'
mail= EmailMessage() #it is an object to EmailMessage CLASS
mail['From'] = sender
mail['To'] = f
mail['Subject'] = subject
mail.set_content(body)
context = ssl.create_default_context()
with smtplib.SMTP_SSL('smtp.gmail.com' , 465,context= context) as mi:
mi.login(sender,paswd)
mi.sendmail(sender,f,mail.as_string())
print(f'mail sent successfully to {f} please check your email inbox')
```

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